Machine-actionable access policies using ODRL

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Abstract
With the coming of the EHDS semantic access policies has become increasingly important for machine-actionable reuse of health data. Since the addition of ODRL in DCAT there is a way within the FDP to facilitate these access policies. Within this paper we propose some functionalities to execute ODRL.

Keywords
ODRL, Semantics, EHDS, FAIR [1]

1. Introduction

With the introduction of the European Health Data Space (EHDS) Regulation every member state within the European Union is now committed to making their health data Findable, Accesible, Interoperable and Reusable (FAIR). The FAIR Data Point (FDP) [1] is a practical implementation for exposure of FAIR metadata using DCAT and DCAT-AP. From DCAT v2 a new standard has been added for machine-actionable accessibility conditions named: Open Digital Rights Language (ODRL). ODRL could become a fundamental building block for the EHDS, but practical implementations of a framework that can use ODRL statements are still lagging behind. Within this paper we propose a framework that can read ODRL statements, check compliance with the statements, and execute them.

2. Data Control

Data sharing generates challenges for the party sharing the data since they lose control over what happens with the data after sharing. Inherent power asymmetry between parties with less and more resources are enhanced by taking actual choice away. Combining a FDP, ODRL, and a data visiting infrastructure these entities with less resources are better able to stay in control over their data by only giving temporary visiting rights to parties to analyze data close to where it is stored. Within ODRL it is possible to draft contracts that are visible, auditable and enforceable by machines. To create a first version of a framework it was needed to define a use-case.

3. Use-case based development of an ODRL framework

The use-case to create the ODRL framework was reused from the Innovative Health Initiative funded PaLaDIn project [2] and was: "*Leo is 17 years old and has DMD. He currently resides in Germany where he lives at home with his parents. He uses a wheelchair and while he has arm movement, it is declining. He wants to enter his data to record the decline in his condition and track this in reference to his care and appointments as well as his daily life and hobbies. He currently plays the piano, but this is proving more difficult as time progresses. For the most part he is interested in his condition but is also interested in the possibility of seeing his data against averages of others with DMD.*"

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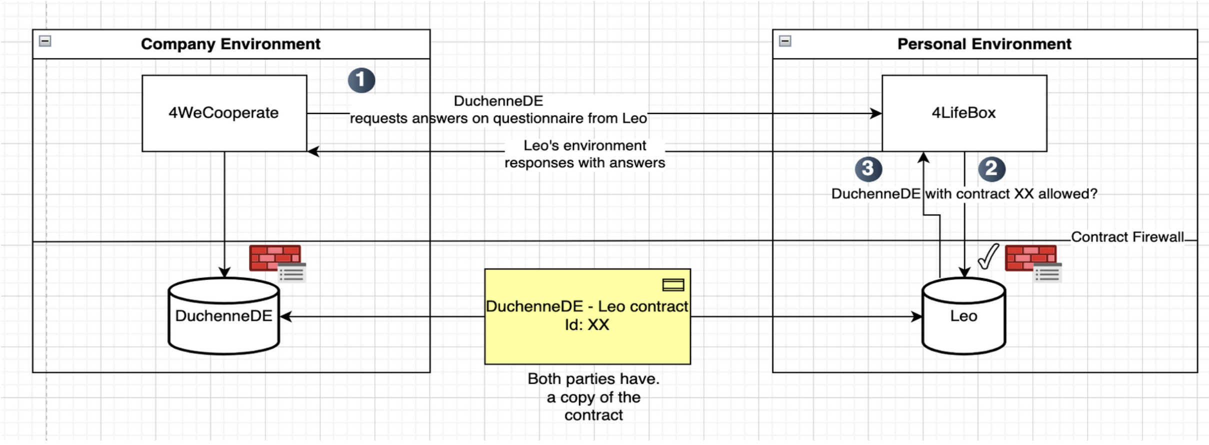

Figure 1: A diagram to show interactions between different parties based on the defined use-case.

In this example, an ODRL contract between DuchenneDE and Leo has already been established and accepted. Both parties have a copy of the contract stored in their own environment. The logic in Figure 1 is as follows: 1) DuchenneDE, who has a company environment, wants to request the answers on their questionnaire from Leo. Their environment asks Leo's environment: I want access to this specific data. 2) DuchenneDE sends the contract ID. Leo's environment checks if that contract exists for him and what data DuchenneDE wants to see and if that's allowed. We coined the term 'Contract Firewall' for this. 3) Leo's environment responds with the questionnaire answer results, since DuchenneDE is allowed to see them.

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Declaration on Generative AI

The author(s) have not employed any Generative AI tools.

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